Truncation preserves finite products: the induced map from the truncation of A x B is an equivalence

The Univalent Foundations Program

hott-book:hlevels.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author's 'cor:trunc-prod' (source lines 610-612, Theorem 7.3.8); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own sentence announcing that the reflector onto n-types preserves finite products, which is what the statement proves for the binary case.

Editorial note. The retained counter values (chapter 7, section 3, theorem offset 7) make the retained theorem print as Theorem 7.3.8, matching the original book numbering.

Editorial note. No retained line refers to a label defined outside the retained span, so the package declares no external reference and no bibliography.

Editorial scope. One original result of the HoTT Book, counted once: truncation preserves finite products (Theorem 7.3.8).

Allowed standard prerequisites (editorial).

- Universal property of n-truncation. For an n-type C, composition with |.|_n is an equivalence from trunc n A -> C to A -> C.. It is applied at source lines 619 and 620 to replace trunc n A -> (B -> C) by A -> (B -> C) and trunc n B -> C by B -> C.
- Function types into an n-type are n-types. If C is an n-type then B -> C is an n-type.. It is invoked at source line 623 so that the universal property applies to trunc n A -> (trunc n B -> C).
- Currying. (A x B -> C) is equivalent to (A -> (B -> C)).. It is the identification used between the first and last lines of the displayed chain.

The category of n-types also has some special properties not possessed by all reflective subcategories. For instance, the reflector $\|-\|_n$ preserves finite products.

Theorem 7.3.8. *For any types A and B, the induced map $\|A \times B\|_n \to \|A\|_n \times \|B\|_n$ is an equivalence.*

Proof. It suffices to show that $\|A\|_n \times \|B\|_n$ has the same universal property as $\|A \times B\|_n$. Thus, let C be an n-type; we have

$$\begin{aligned}(\|A\|_n \times \|B\|_n \to C) &= (\|A\|_n \to (\|B\|_n \to C)) \\ &= (\|A\|_n \to (B \to C)) \\ &= (A \to (B \to C)) \\ &= (A \times B \to C)\end{aligned}$$

using the universal properties of $\|B\|_n$ and $\|A\|_n$, along with the fact that $B \to C$ is an n-type since C is. It is straightforward to verify that this equivalence is given by composing with $|-|_n \times |-|_n$, as needed. □